



The role of artificial intelligence in early detection and risk prediction of ischemic heart disease

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Abstract

Ischemic heart disease (IHD) remains a leading cause of global morbidity and mortality, underscoring the need for rapid and accurate diagnostic strategies. Conventional methods, including electrocardiography (ECG), imaging, and biomarkers, are effective but limited by factors such as delayed biomarker elevation, reliance on expert interpretation, and variability across settings. Artificial intelligence (AI) offers new opportunities to enhance early detection and risk prediction by applying machine learning and deep learning to large, complex datasets.

In ECG analysis, AI models consistently identify subtle ischemic patterns, including occlusive myocardial infarction, with accuracy that often rivals or exceeds clinicians. In imaging, AI enhances echocardiography, CT, MRI, and nuclear modalities by automating segmentation, strain analysis, and plaque quantification while reducing interpretation time. In biomarkers, AI augments traditional tools like troponins and enables the discovery of novel predictors through multi-omics and wearable data integration, supporting dynamic and individualized risk assessment.

Despite promising results, most studies remain retrospective or single-center, with limited validation across diverse populations and healthcare environments. Key barriers include algorithm bias, generalizability, regulatory uncertainty, and limited clinician familiarity. Future progress will depend on multicenter trials, federated learning, explainable AI, and integration into existing workflows.

In conclusion, AI has the potential to transform cardiovascular care by enabling earlier and more precise diagnosis of IHD and more personalized risk prediction. However, realizing this potential will require careful validation, equitable implementation, and collaboration across disciplines to ensure safe and effective adoption in clinical practice.

Keywords: AI, artificial intelligence, biomarkers, cardiac imaging, CVD, ECG, IHD, ischemic heart disease, MI, myocardial infarction, risk prediction, acute, A, myocardial infarction, MI

Introduction

Ischemic Heart Disease (IHD), commonly known as a heart attack, remains one of the leading causes of morbidity and mortality worldwide^[1]. Rapid and accurate diagnosis is critical to improving patient outcomes, as early intervention significantly reduces complications and mortality rates^[2]. Traditionally, IHD is diagnosed using a combination of electrocardiography (ECG), cardiac biomarkers (such as troponin levels), and imaging techniques (echocardiography, coronary

HIGHLIGHTS

- AI enables early and accurate diagnosis of ischemic heart disease.
- It improves ECG, imaging, and biomarker-based risk prediction.
- AI models outperform traditional tools in speed and precision.
- Bias and poor validation limit AI use across diverse patients.
- Clinicians' trust and strong regulation are key to AI adoption.

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angiography, and cardiac MRI). However, these conventional methods have limitations, including delays in biomarker elevation and dependency on specialized personnel and equipment^[3–5]. Artificial intelligence (AI) has emerged as a transformative tool in modern cardiology, offering the potential to enhance the early detection of IHD^[6,7]. AI-driven models, particularly machine learning (ML) and deep learning algorithms, can analyze complex patterns in ECG data, imaging scans, and electronic health records with superior speed and accuracy^[8–10]. Recent studies have demonstrated that AI can outperform traditional diagnostic methods in identifying IHD, enabling faster decision-making and reducing diagnostic errors^[11,12].

Recent reviews and consensus statements have made significant contributions to the field of AI in cardiovascular medicine; however, their focus has generally been broader or modality-specific, with less emphasis on IHD. For example, Schulze *et al* focused on expert consensus for CCTA-based plaque evaluation^[13]. In contrast, others have concentrated narrowly on technical aspects of echocardiography^[14,15] or modality-specific advances in CT, MRI, and nuclear imaging^[16–19]. Broader surveys^[20,21] have outlined AI's potential across the cardiovascular spectrum, but without a detailed synthesis of ischemia-specific evidence. This review explores the role of AI in the early detection and management of IHD, with a particular focus on acute myocardial infarction (AMI). Unlike prior reviews, it provides a disease-centered, cross-modality perspective that complements previous work by integrating evidence from ECG interpretation, cardiac imaging, biomarker analysis, and risk prediction models. Additionally, we discuss the benefits and challenges of clinical implementation as well as future research directions.

Traditional diagnostic and predictive methods

The diagnosis of AMI relies on a combination of clinical evaluation, ECG findings, biomarker analysis, and imaging studies. However, each of these methods presents limitations that can delay or complicate early detection^[22,23].

ECG is one of the first-line diagnostic tools used for AMI detection. ST-segment elevation myocardial infarction (STEMI) and non-STEMI (NSTEMI) are classified based on ECG changes^[24]. However, ECG interpretation is subject to variability due to differences in physician expertise and non-specific findings in some cases^[25]. Echocardiography and coronary angiography provide essential structural and functional information about the heart. While useful, these methods require specialized equipment and trained personnel, making them less accessible in resource-limited settings^[26,27]. Cardiac troponins (cTnI and cTnT) are the gold standard biomarkers for AMI diagnosis^[28]. However, they require several hours to rise to detectable levels, leading to potential delays in treatment initiation^[29].

The ASCVD risk score predicts the 10-year risk of heart attack or stroke based on factors like age, cholesterol, and blood pressure^[30]. However, it has limitations in predicting risks for individuals outside the average population. AI innovations improve prediction by integrating dynamic data such as genetic information and real-time biomarkers, providing more personalized and accurate cardiovascular risk assessments^[31,32]. Given these challenges, AI-driven solutions are being developed to enhance the accuracy and efficiency of AMI detection, particularly in ECG interpretation, echocardiographic analysis, cardiac biomarker assessment, and risk stratification.

Methods

This review was conducted using a structured narrative methodology, combining elements of systematic searching with thematic qualitative synthesis. The objective was to summarize and integrate current evidence on AI applications for the early detection and risk prediction of ischemic heart disease (IHD). Because the included studies represent a heterogeneous mix of modalities (ECG, imaging, biomarkers, prediction models), study designs,

and validation approaches, a full systematic review with meta-analysis was not methodologically appropriate.

Search sources and databases

A structured search was performed in the following databases: PubMed/MEDLINE, Embase, Scopus, and Web of Science. The search included articles published from database inception to September 2025, restricted to English-language studies.

Search strategy

Searches used combinations of controlled vocabulary and free-text terms related to:

“artificial intelligence,” “machine learning,” “deep learning”
 “electrocardiography,” “ECG,” “echocardiography,” “cardiac CT,” “MRI,” “nuclear cardiology”
 “biomarkers,” “troponin,” “omics,” “wearable devices”
 “ischemic heart disease,” “coronary artery disease,” “myocardial infarction”

An example PubMed search string was: (“artificial intelligence” OR “machine learning” OR “deep learning”) AND (“ischemic heart disease” OR “coronary artery disease” OR “myocardial infarction”) AND (electrocardiography OR ECG OR echocardiography OR “computed tomography” OR CT OR MRI OR “CT angiography” OR “computed tomography angiography” OR “coronary CT angiography” OR “CTA” OR “CT angio” OR “cardiac imaging” OR biomarkers OR troponin OR “risk prediction” OR “risk stratification” OR angiography OR “coronary angiography” OR PET OR “positron emission tomography”). Search strategies for other databases used analogous terms.

Eligibility approach

Because the purpose was to produce a cross-modality, clinically oriented synthesis, inclusion decisions were based on relevance rather than exhaustiveness. Studies were included if they: Examined AI or ML applications relevant to IHD detection, prediction, or risk stratification, Reported original data or represented high-impact consensus/guideline papers, Provided evaluable outcomes (diagnostic accuracy, AUC, workflow improvement, biomarker performance, etc.).

Exclusion criteria included: Non-human studies, Conference abstracts without full data, Papers focusing exclusively on model development without clinical application, Studies irrelevant to ischemia

Selection process

Two authors independently screened titles and abstracts. Full texts of potentially relevant articles were reviewed, and disagreements were resolved by consensus. Additional key articles were identified through citation tracking of major reviews, guidelines, and landmark studies.

Synthesis approach

Given the heterogeneity of AI applications across different modalities, a qualitative, domain-specific synthesis was performed rather than quantitative pooling. Studies were grouped into four predefined domains: (1) ECG-based AI, (2) AI in cardiac imaging, (3) AI in biomarkers and multi-omics, and (4) AI in risk

prediction models. Structured summary tables were created to highlight study design, modality, sample size, validation methods, and main outcomes.

AI in electrocardiography

Diagnostic applications in AI-ECG

The integration of artificial intelligence (AI) into electrocardiogram (ECG) interpretation represents a groundbreaking advancement in cardiovascular medicine, particularly for the detection of AMI. By harnessing sophisticated deep learning architectures such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and hybrid models, AI-ECG systems can identify subtle, clinically significant patterns that often elude human interpreters. This capability was powerfully demonstrated by Attia *et al*, whose landmark study showed AI could detect asymptomatic left ventricular dysfunction (LVEF $\leq 35\%$) from routine ECGs with an AUC of 0.93, proving ECG signals contain far more diagnostic information than traditionally appreciated^[33]. These capabilities enable earlier and more precise AMI diagnosis, fundamentally transforming emergency cardiac care and preventive cardiology^[34–37].

AI models have demonstrated remarkable superiority over conventional rule-based algorithms and even experienced clinicians in detecting STEMI. Ribeiro *et al* provided crucial multicenter validation of this potential, showing AI could diagnose 24 cardiac pathologies across international populations with expert-level accuracy (average AUC: 0.97)^[38]. Controlled studies consistently report sensitivity and specificity exceeding 95%, with some architectures achieving near-perfect accuracy. For example, a specialized CNN trained on the PTB database attained an unprecedented 99.1% accuracy in MI detection using only three-lead ECGs, while advanced multi-scale ResNet models with attention mechanisms reached 99.98% specificity when localizing AMI^[35–37,39]. These performance metrics suggest AI could soon become the gold standard for initial ECG interpretation.

Beyond STEMI: detecting occlusive MI with ECG

While traditional STEMI criteria remain foundational, they fail to identify approximately 30% of acute coronary occlusions, potentially delaying life-saving interventions. Modern AI systems address this critical gap by incorporating recognition of non-ST elevation patterns such as hyperacute T-waves and de Winter's T-waves. The FDA's 2023 guidance on AI/ML-based software specifically highlights the need for such advanced detection capabilities, emphasizing algorithms that identify life-threatening conditions beyond binary STEMI/non-STEMI classifications^[40]. This expanded capability allows for earlier detection of occlusive myocardial infarction (OMI), with proprietary algorithms like Queen of Hearts demonstrating a 34% reduction in false-positive cath lab activations while maintaining perfect sensitivity for true OMI cases^[41,42]. Such advancements are reshaping clinical protocols for acute coronary syndrome management.

AI-ECG in prehospital triage and acute care

The implementation of AI in prehospital settings has shown particular promise through tiered risk stratification systems. By analyzing multiple parameters, including heart rate, QRS width,

and signal artifacts, these algorithms can effectively categorize ECGs into low, intermediate, and high-risk groups. This approach has reduced unnecessary physician overreads by 62% without compromising sensitivity, significantly streamlining emergency cardiac workflows^[43,44]. Furthermore, the development of smartwatch-compatible AI models capable of asynchronous multi-lead analysis (achieving AUROC scores of 0.845 for 3-lead ECGs) opens new possibilities for remote patient monitoring and community-based screening^[45,46].

Expanded applications of AI-ECG beyond ischemic disease

Recent work shows AI-ECG extending beyond ischemia into structural disease screening, prognosis, arrhythmia prediction, and equity. Poterucha *et al* (2025)^[47] applied a multitask CNN (EchoNext) to >1.2 million ECG–echo pairs, detecting composite structural heart disease with AUROC 0.85 internally and ~0.78–0.80 externally, outperforming cardiologists though still limited by retrospective design. Sau *et al* (2024)^[48] trained a survival CNN (AIRE) on >1.1 million ECGs, with multinational validation showing robust prediction of mortality and heart failure (C-index 0.64–0.78), suggesting scalable prognostic use. Sue *et al* (2025)^[49] analyzed >247 000 ambulatory ECGs from six countries, predicting sustained VT within 13 days with AUROC ~ 0.95, highlighting potential for wearable-based arrhythmia forecasting. Jain *et al* (2023)^[50] combined ECG and echo (ATTRACTnet) to detect transthyretin amyloidosis (AUROC: 0.90) while enforcing fairness constraints, achieving similar accuracy across Black and White patients. Together, these studies broaden AI-ECG's remit but remain largely retrospective, requiring prospective trials to confirm clinical impact.

Technical innovations in AI-ECG analysis

Signal processing and feature extraction: Cutting-edge approaches that combine Tunable-Q Wavelet Transform (TQWT) with variational mode decomposition (VMD) have enabled the extraction of previously inaccessible diagnostic information from ECG and VCG signals. These hybrid methods have achieved 98.2% accuracy in MI detection without relying solely on ST-segment changes, representing a significant departure from conventional interpretation paradigms^[46,51]. Such innovations demonstrate how AI can uncover novel biomarkers within standard cardiac recordings.

AI interpretability and clinical integration, enhancing trust and workflow: Modern AI systems increasingly incorporate explainability features to facilitate clinician trust and adoption. Attention mechanisms like Grad-CAM visually demonstrate lead-specific contributions to the algorithm's decision-making process, effectively aligning AI outputs with established ECG-topography principles^[37]. Additionally, the implementation of certainty indices has proven effective in mitigating automation bias, with studies showing a 15% improvement in clinician diagnostic accuracy when these indices accompany AI interpretations^[34]. These developments address longstanding concerns about AI's "black box" nature while optimizing human-AI collaboration.

Challenges and the future of AI in ECG

Despite their impressive capabilities, AI-ECG systems face significant challenges related to demographic biases in training data. Performance disparities across age, sex, and racial groups

currently limit real-world applicability, underscoring the need for more diverse datasets. Initiatives like the 7.1-million-patient cohort study^[52] represent crucial steps toward developing equitable algorithms that perform consistently across all patient populations.

The path to widespread clinical implementation remains obstructed by several factors, including the persistent “black box” problem inherent to many complex algorithms. Recent FDA guidance (2023) establishes critical frameworks for continuous monitoring of AI performance across diverse demographics, addressing a key limitation identified in studies like Ribeiro *et al.*, which showed geographic variability in algorithm accuracy^[40]. While rhythm-detection AI has obtained FDA clearance, more advanced systems for OMI and structural heart disease detection remain in investigational stages^[53]. Overcoming these barriers will require continued improvements in algorithmic transparency, rigorous multicenter validation, and the development of standardized regulatory pathways.

The future of AI-ECG technology extends far beyond simple detection algorithms. Building on Attia *et al.*'s discovery of ECG biomarkers for LV dysfunction, newer models now predict three-year acute coronary syndrome risk (hazard ratio 11.75 for high-risk ECG phenotypes)^[33]. Integration with wearable devices promises to enable continuous AMI risk monitoring while prognostic applications show remarkable potential for predicting left ventricular dysfunction (AUC: 0.93) and 3-year acute coronary syndrome risk (hazard ratio 11.75 for high-risk ECG phenotypes)^[52,54]. AI-ECG systems represent a paradigm shift in the approach to AMI diagnosis, combining unprecedented analytical capabilities with the potential for widespread deployment. These developments suggest AI-ECG may soon transition from a diagnostic tool to a comprehensive cardiovascular risk assessment platform. As the technology matures, the coming years will likely see AI-ECG become an indispensable component of cardiovascular care, fundamentally changing how we detect and manage acute coronary syndromes.

As summarized in Table 1, AI-ECG studies focused on ischemic heart disease^[34–38,42,43] consistently demonstrate strong diagnostic potential. However, performance varies between curated datasets, which often achieve near-perfect accuracy, and larger, real-world cohorts, which show more modest but clinically meaningful results. Beyond IHD, models also extract signals for structural disease^[33,47], prognosis^[48], and arrhythmia forecasting^[49,55], with fairness-aware approaches emerging^[50]. Points of divergence lie mainly in methodology: curated datasets tend to overestimate accuracy, while multicenter cohorts reveal variability in generalizability and calibration. Direct comparisons with clinicians are mixed; in some contexts, AI outperforms, in others, physicians remain superior. Unresolved questions remain about real-world impact: Do these models improve patient outcomes when deployed prospectively? Can they generalize across populations, devices, and vendors? How should prevalence and fairness constraints be managed in practice? Addressing these gaps will require prospective multicenter trials, routine external validation, and standardized reporting of calibration and equity metrics.

AI in medical imaging

Diagnostic applications in AI-imaging

Echocardiography, enhancing precision and speed: The integration of AI into cardiac imaging represents a paradigm shift in

how we detect and manage AMI, as recognized in recent consensus guidelines from the American College of Cardiology^[57]. Across echocardiography, cardiac CT, MRI, and nuclear imaging, AI algorithms are demonstrating remarkable capabilities that address longstanding diagnostic challenges while creating new opportunities for precision medicine, mirroring trends seen across medical specialties^[58].

Cardiac CT, anatomical and functional insights: In echocardiography, AI has revolutionized strain imaging by automating global longitudinal strain (GLS) measurements with significantly better reproducibility ($\kappa = 0.82$) compared to conventional methods ($\kappa = 0.68$)^[59]. This advancement is particularly valuable for detecting subtle ischemic dysfunction in complex cases like chemotherapy-induced cardiotoxicity. The clinical impact extends to emergency settings, where AI-powered handheld devices now enable rapid AMI triage, reducing time-to-diagnosis from 42 to 18 minutes in recent trials^[60], consistent with findings from the multicenter AI-ECHO registry^[56]. Comprehensive AI models that integrate multiple echocardiographic parameters have achieved exceptional diagnostic accuracy (AUC: 0.96) for AMI detection^[61], while also identifying complications like ventricular thrombus with 94% sensitivity^[61].

Cardiac CT has undergone perhaps the most dramatic transformation through AI. Advanced plaque analysis algorithms now automatically quantify high-risk features like a low-attenuation plaque with 91% accuracy^[16,62], outperforming traditional risk stratification methods. The development of AI-derived fractional flow reserve (FFR-CT) has been particularly impactful, showing 92% agreement with invasive FFR while reducing unnecessary angiography by 38%^[16,63], results corroborated by the multicenter NXT trial and the recent randomized controlled trial of an AI-based CT-FFR algorithm^[64,65]. Recent systematic reviews highlight that a wide array of AI techniques – including machine learning and deep learning models – can non-invasively estimate FFR, offering diagnostic performance comparable to invasive approaches and supporting clinical decision-making while mitigating procedural risks^[66]. Moreover, emerging deep-learning models such as FFR-AI have demonstrated diagnostic accuracy of 85% in predicting lesion-specific ischemia ($\text{FFR} \leq 0.80$), performing comparably to traditional computational fluid dynamics-based FFRCT, but with potentially faster computation and reduced complexity^[67]. Opportunistic screening represents another breakthrough, with AI extracting coronary calcium scores from non-gated CTs with 95% concordance to dedicated scans^[68] – a capability that could transform preventive cardiology if implemented widely.

MRI and Nuclear imaging, advances in structural and perfusion analysis: Cardiac MRI benefits from AI through enhanced tissue characterization. Automated late gadolinium enhancement (LGE) analysis reduces interobserver variability in infarct sizing by 40–60% compared to manual methods^[69], while multiparametric models integrating cine, T1/T2 mapping, and perfusion data improve detection of subtle AMI patterns (sensitivity increases from 72% to 89%)^[70]. These advancements are particularly valuable for assessing myocardial recovery potential post-revascularization (AUC: 0.87)^[70], building on earlier radiomics work predicting microvascular obstruction^[71].

In nuclear cardiology, AI improves both diagnostic accuracy and workflow efficiency. Machine learning enhances SPECT/PET interpretation (accuracy improvement from 82% to 91%)

Table 1
Artificial intelligence in ECG interpretation: current applications and research insights.

AI in ECG interpretation								
Study (author, year)	Population/dataset	Data modality	Purpose	AI method	Validation	Key performance	Key findings	Limitations
Poterucha <i>et al.</i> , 2025 ^[47]	1.24 M ECG-TTE pairs; external sets from three centers	12-lead ECG + demographics	Detect structural heart disease	Multitask CNN	Large internal + external; small prospective sample	AUROC internal 0.85; external 0.78–0.80	Outperformed cardiologists; good generalizability	Prospective sample small; PPV prevalence-dependent
Sau <i>et al.</i> , 2024 ^[48]	1.16 M ECGs + 4 multinational cohorts	12-lead ECG ⁶⁶	Predict mortality, HF, arrhythmias	CNN survival model	Multinational external (retrospective)	C-index 0.64–0.78	Strong prognostic value; biologically plausible	ICD-based labels; no prospective validation
Ribeiro <i>et al.</i> , 2020 (CODE) ^[38]	2.3 M ECGs; annotated test set	12-lead ECG	Diagnose 6 ECG abnormalities	1-D ResNet	Internal validation	F1 > 80%; Spec > 99%	High performance at scale	Limited to 6 diagnoses; label noise possible
Attia <i>et al.</i> , 2019 ^[33]	~97K ECG-echo pairs (Mayo Clinic)	12-lead ECG	Detect low EF (≤35%)	CNN	Large internal validation	AUC: 0.93	High NPV; predicts future LV dysfunction	Single-center; PPV modest; no external testing
Florina <i>et al.</i> , 2025 ^[48]	247K ambulatory ECGs from six countries	Single-lead ambulatory ECG	Predict near-term VT	Multi-branch deep model	Internal + external	AUROC 0.95–0.97	Strong near-term VT prediction	Few positive cases; retrospective only
Jain <i>et al.</i> , 2023 ^[21]	~9k paired ECG/TTE + PYP scans	ECG + echo metrics	Detect ATTR amyloidosis	ATTRACTnet (ECG + TTE)	Retrospective	AUROC 0.90	Outperformed Mayo score; reduced racial bias	Single-center; no external testing
Baker <i>et al.</i> , 2025 ^[42]	EMS cohort (117 suspected OMI)	Prehospital 12-lead ECG	Reduce false cath lab activations	Proprietary CNN	Retrospective	↓ false positives by 34%	Detected all true OMIs	Small single-center cohort
Wu <i>et al.</i> , 2023 ^[38]	9340 ED ECGs	12-lead ECG	Compare CIE vs physician STEMI reads	Philips DXL (rule-based)	Retrospective	Physicians: AUC: 0.94; DXL: 0.86	Human reads more accurate	Single-center; non-ML model
He <i>et al.</i> , 2025 ^[58]	MIT-BIH arrhythmia dataset	ECG beat segments	Classify 5 arrhythmias	CardioAttentionNet	5-fold CV + held-out test	ACC ~ 94%; AUC: 97–99%	Low-latency, wearable-ready model	No clinical external validation
Cao <i>et al.</i> , 2022 ^[37]	PTB: 682k beats	12-lead beat-level ECG	Detect + localize MI	Multi-scale ResNet with attention	Retrospective (public dataset)	Acc >99%	Excellent performance on clean data	Limited real-world generalizability
Makimoto <i>et al.</i> , 2020 ^[98]	PTB Diagnostic (289 ECGs)	ECG images (PNG)	Detect MI	CNN (image-based)	Internal testing	F1 83%; Acc 81%	Outperformed physicians	Very small dataset; static images
Kroery <i>et al.</i> , 2019 (SPICED-ACS) ^[34]	3262 ECG reads (survey)	12-lead ECG	Assess effect of “certainty index”	Rule-based certainty index	Survey (not patient data)	OR 1.06 for improved accuracy	Decision support may help readers	Not a diagnostic study; artificial index
Goebel <i>et al.</i> , 2019 ^[43]	63 prehospital STEMI-flagged ECGs	12-lead ECG	Stratify probability tiers	Rule-based algorithm	Small prospective	PPV 100% in high-probability tier	Reduced unnecessary reviews	Very small sample; vendor-specific
Acc: accuracy; AUC: accuracy; AF: atrial fibrillation; AI: artificial intelligence; ASCVD: atherosclerotic cardiovascular disease; ATTR-CA: transthyretin cardiac amyloidosis; AUPRC: area under the receiver operating characteristic curve; AV block: atrioventricular block; BLSTM: bidirectional long short-term memory; CCL: cardiac catheterization laboratory; CI: confidence interval; CIE: computer-interpreted eeg; CNN: convolutional neural network; DISCOVERY trial: name of prospective trial for ai recruitment; DNN: deep neural network; ECG: electrocardiography; ECGNet: ai model name; ED: emergency department; EMS: emergency medical services; F1: F1 score; FC layers: fully connected layers; Grad-CAM: Gradient-weighted Class Activation Mapping; GWAS: Genome-Wide Association Study; HF: heart failure; HR: heart rate; LBBB: left bundle branch block; LVEF: left ventricular ejection fraction; LVH: left ventricular hypertrophy; NPV: negative predictive value; OMI: occlusion myocardial infarction; PhelWAS: Phenome-Wide Association Study; PNG: portable network graphics; PPV: positive predictive value; PREC: precision; PTB DB: Physikalisch-Technische Bundesanstalt ECG Database; PVC: premature ventricular contraction; RBBB: right bundle branch block; REC: recall; RV: right ventricle; SB: sinus bradycardia; SENet: squeeze-and-excitation network; Sens: sensitivity; SHD: structural heart disease; Spec: specificity; SPICED-ACS: study name for stemi interpretation survey; ST: ST-segment deviation; STEMI: ST-elevation myocardial infarction; TCAS: transthyretin cardiac amyloidosis score; TTE: transthoracic echocardiography; VA: ventricular arrhythmia; VT: ventricular tachycardia.								

while reducing reading time by 35%^[72]. Perhaps more importantly, AI models combining perfusion data with clinical variables outperform traditional risk scores for predicting major adverse cardiac events (AUC: 0.92 vs 0.78) and appropriately reclassify 31% of intermediate-risk patients^[72,73].

Technical innovations in AI-imaging

The technical innovations enabling these clinical applications are equally noteworthy. Radiomics analysis extracts hundreds of sub-visual features from cardiac images, identifying high-risk plaque characteristics (AUC: 0.88)^[74] and predicting microvascular obstruction (AUC: 0.91)^[75] with precision beyond human capability benchmarks^[76]. Workflow automation through AI has reduced echocardiogram interpretation time by 48% while maintaining diagnostic accuracy^[61], and intelligent triage systems prioritize urgent cases in PACS, cutting time-to-diagnosis by 25–40 minutes^[60,61]. Explainable AI (XAI) methods address the “black box” problem by generating visual explanations of AI decisions^[77] and providing confidence scores that improve clinical trust^[78].

Challenges and the future of AI in imaging

However, significant challenges remain. Most AI models in cardiovascular imaging are trained on limited, single-center datasets^[70,78], and systematic reviews show that their performance commonly declines on external validation – ranging from modest reductions to very large drops exceeding 80% in some studies^[79,80]. Only 5% of studies have tested AI in real clinical settings^[70], and just four AI tools have received FDA approval for cardiac applications^[73]. Workflow integration presents another hurdle, with 68% of AI systems operating as standalone rather than PACS-integrated solutions^[68]. Perhaps most importantly, 72% of cardiologists lack formal AI training^[81], creating a knowledge gap that must be addressed for successful implementation.

The path forward requires multicenter collaborations to develop diverse training datasets, prospective trials demonstrating real-world impact, and standardized guidelines for clinical integration. Professional societies are already addressing these needs through initiatives like the RSNA Common Data Elements project and ACC AI in Cardiology working groups. As these efforts progress, AI is poised to transition from an assistive tool to a fundamental component of cardiac imaging workflows, following the implementation roadmap proposed by the NIH Medical Imaging and Data Resource Center.

AI for cardiac imaging has produced several ischemia-relevant advances alongside broader cardiac applications, as outlined in Table 2. For IHD, CCTA radiomics^[71] produced a perivascular Fat Radiomic Profile that improved MACE prediction beyond stenosis and calcium ($\Delta C\text{-stat} + 0.126$; adjusted HR = 10.8) and was validated in SCOT-HEART, while AI approaches to non-invasive lesion ischemia^[67,82] and CT-FFR show high diagnostic accuracy (AUCs = 0.80–0.98) but are largely single-center and often require manual edits. Cine MRI → synthetic LGE^[83] showed strong scar detection (Sens = 91%, Spec = 96%) and good scar-size correlation, suggesting a path to contrast-free infarct imaging. By contrast, SPECT radiomics^[84] yielded modest diagnostic performance but useful risk stratification (risk AUC = 0.79). Outside IHD, automated echo segmentation and EF estimation^[85–87] achieve excellent reproducibility and

accuracy (Dice > 0.9; EF $r = 0.96$; AUC = 0.97 for reduced EF), and CMR/LGE classifiers^[88] report very high single-center accuracy but lack broad external testing. Common themes and disagreements are methodological: small, curated single-site studies frequently report near-perfect metrics, whereas larger multicenter validations report lower but more realistic performance; many high-accuracy reports depend on specific vendors/scanners, manual correction steps, limited disease spectra, or small numbers of clinical events. Key remaining gaps are therefore practical and translational: prospective multicenter impact trials, rigorous external validation across vendors/populations, automation of preprocessing/segmentation, standardization of ground-truth labels, adequate event counts for prognostic claims, and formal assessment of clinical utility, workflow integration, and cost-effectiveness. These steps are necessary before widespread clinical deployment.

In conclusion, AI is transforming AMI detection across all cardiac imaging modalities, offering improvements in diagnostic accuracy, workflow efficiency, and prognostic capability that were unimaginable just a decade ago^[93]. From automated strain analysis in echocardiography to AI-driven plaque characterization in CT and precision tissue assessment in MRI, these technologies are redefining our approach to ischemic heart disease. While challenges remain in implementation and validation, the potential benefits for patient care are enormous. With continued innovation and collaboration, AI will undoubtedly become an indispensable tool in the fight against AMI, improving outcomes for millions of patients worldwide.

AI in biomarker discovery/analysis

Enhancing traditional biomarkers with AI

As our understanding of cardiovascular diseases advances, so does our ability to detect and manage them through biomarkers. The integration of artificial intelligence (AI) into biomarker research and clinical practice is driving unprecedented innovations in the prediction, diagnosis, and monitoring of cardiovascular health. Artificial intelligence is breathing new life into conventional cardiac diagnostics by extracting unprecedented value from existing tests. For ECGs, AI algorithms like the QCG™ system don't just look for ST elevations – they analyze subtle waveform morphologies, repolarization patterns, and dynamic changes across leads to detect coronary occlusions with 77% accuracy (AUC: 0.770), significantly outperforming physician interpretation^[94]. This is particularly valuable for identifying non-traditional ischemia patterns like hyperacute T-waves or de Winter's signs that often get missed in emergency settings. Similarly, for troponins, AI goes beyond simple threshold detection – machine learning models examine the nuanced kinetics of enzyme release, contextualizing absolute values with the patient's baseline characteristics, timing of symptoms, and concurrent conditions to dramatically reduce false positives while maintaining high sensitivity^[95]. These AI-enhanced interpretations are making our oldest cardiac tests smarter and more reliable.

Beyond traditional biomarkers: discovering novel biomarkers with AI

The biomarker revolution is being supercharged by AI's ability to find meaningful signals in complex biological data. Multi-

Table 2
Artificial intelligence in medical imaging: current applications and research insights.

AI in Medical Imaging								
Study (author, year)	Population/dataset	Data modality	Purpose	AI method	Validation	Key performance	Key findings	Limitations
Okonomou <i>et al.</i> , 2019 ^[71]	167 biopsies; 202 CCTA (MACE vs controls); SCOT-HEART (n = 1575)	CCTA + PVAT	Predict MACE using fat radiomics	Random Forest	Internal + large external	↑ C-stat + 0.126; HR 10.8 for high FRP	FRP predicted MACE beyond stenosis/CAC	Retrospective; slow processing; LCx not assessed
Giannopoulos <i>et al.</i> , 2023 ^[62]	59 pts (74 lesions) with invasive FFR/FFR	Coronary CTA	Predict lesion-level ischemia	Multichannel U-Net + DL pressure drop	Single-center	AUC: 0.98; Acc 96%	High accuracy even in calcified lesions	Small dataset; manual edits; single vendor
Peters <i>et al.</i> , 2024 ^[67]	37 pts; 39 stenoses	CCTA multiplanar reconstructions	Ischemia prediction vs CFD	InceptionV3 CNN	Single-center	Sens 91%; Spec 82%	Comparable/better than CFD-FFRCT	Only 39 lesions; binary cutoff; manual steps
Amini <i>et al.</i> , 2023 ^[64]	395 SPECT MPI pts	Rest/stress SPECT	Diagnose CAD; risk stratification	Radiomics + ML	Single-center	AUC: 0.61 (dx), 0.79 (risk)	Diabetes + texture features predictive	No external validation; class imbalance
Qi <i>et al.</i> , 2024 ^[63]	430 MI pts (3 centers) + public sat	Cine CMR	Generate LGE-equivalent images	GAN + 3D CNN + LSTM	Internal + external	Sens 91%; Spec 96%	High-quality synthetic LGE; good scar detection	Philips 3 T only; weak for small scars
Ben Khalifa <i>et al.</i> , 2025 ^[66]	163 pts (MI, myocarditis, healthy)	CMR LGE	Classify MI vs myocarditis vs healthy	VGG16 + attention	Single-center	Acc 96%; AUC ≈ 1.00	Outperformed radiologists	Retrospective; no external validation
Brave <i>et al.</i> , 2024 ^[69]	71,589 internal + 8,003 external CXRs	PA chest X-ray	Detect LVH and dilation	DenseNet-121	Large internal + external	AUROC 0.76–0.80	Outperformed radiologists; localized LV	Retrospective; echo labels variable
Liu <i>et al.</i> , 2021 ^[66]	340 internal; CAMUS (500); EchoNet (10k)	2D echo	Automated LV segmentation & EF	DPS-Net (U-Net + FPN)	Multicenter external	Dice >0.92, r = 0.96 EF	High accuracy and reproducibility	Retrospective; limited arrhythmias
Vega <i>et al.</i> , 2024 ^[68]	441 outpatient echo pts	2D echo (A2C/A4C/PLAX)	Automated EF & EF<50% detection	ExoAI proprietary	Multicenter outpatient	Acc 80–91%; r = 0.75–0.89	Robust across views; good EF<50% detection	Not MRI-validated; missing views
Kim <i>et al.</i> , 2023 ^[60]	83 subjects (2 centers)	CMR T2 maps	LV segmentation & edema detection	Commercial U-Net	Small two-center	DSC 0.84; Acc 83–92%	Automated T2 non-inferior to manual	Small dataset; Siemens 3 T only
Kalapos <i>et al.</i> , 2023 ^[61]	262 pts; 1,358 T1/T2 maps	Native CMR T1/T2	LV segmentation	U-Net + ResNet-50	Single-center	Dice 86%	Fast inference; accurate T1	Mostly male/Caucasian; single vendor
Xu <i>et al.</i> , 2023 ^[67]	275 MRI scans	Cine CMR	LV/RV/myocardium segmentation	Improved U-Net	Single-center	Dice 0.86–0.97	Good segmentation performance	No external validation; poor apex/base
Lee <i>et al.</i> , 2024 ^[62]	1,432 ED pts, 1,833 CXR–EHR pairs	CXR + EHR multimodal	Predict abnormal NT-proBNP	DenseNet-121 + MLP	Internal CV	AUROC 0.986	Near-instant AHF screening	No external validation; multimodal complexity

2D echo: two-dimensional echocardiography; A2C: apical 2-chamber view; A4C: apical 4-chamber view; Acc: accuracy; AHF: acute heart failure; AI: artificial intelligence; AMI: acute myocardial infarction; AUROC: area under the receiver operating characteristic curve; CAD: coronary artery disease; CCTA: coronary computed tomography angiography; CFD: computational fluid dynamics; Cine MRI: motion MRI; CMR: cardiac magnetic resonance; CNN: convolutional neural network; CTA: computed tomography angiography; CV: cross-validation; CXR: chest X-ray; DenseNet-121–121-layer Dense Convolutional Network; DL: deep learning; DLV: dilated left ventricle; DPS-Net: deep pyramid segmentation network; DSC: dice similarity coefficient; EF: ejection fraction; EHR: electronic health record; FFR: fractional flow reserve; FFRCT: FFR derived from CT; FPN: feature pyramid network; GAN: generative adversarial network; GB: gradient boosting; HSR: heart size ratio; ICC: intraclass correlation coefficient; iFR: instantaneous wave-free ratio; KNN: K-nearest neighbors; LayerCAM: layer-wise class activation mapping; LCx: left circumflex artery; LGE: late gadolinium enhancement; LSTM: long short-term memory; LV: left ventricle; MACE: major adverse cardiovascular events; MLP: multi-layer perceptron; Myo: myocardium; NT-proBNP: N-terminal pro-B-type natriuretic peptide; PA: posteroanterior; PLAX: parasternal long-axis view; Prec: precision; PVAT: perivascular adipose tissue; r: pearson correlation coefficient; ResNet-50: 50-layer residual network; RF: random forest; RMSE: root mean squared error; ROI: region of interest; RV: right ventricle; Sens: sensitivity; SHAP: SHapley Additive exPlanations; SLVH: suspected left ventricular hypertrophy; Spec: specificity; SPECT MPI: single photon emission computed tomography myocardial perfusion imaging; T1/T2 maps: relaxation time maps in MRI; U-Net: convolutional neural network architecture for segmentation; VGG16: 16-layer visual geometry group CNN.

omics analysis represents a major frontier – machine learning can sift through thousands of genomic variants, proteomic signatures (like GDF-15's association with myocardial stress), and metabolomic profiles to identify novel combinations that predict AMI risk long before symptoms appear^[96,97]. Wearable technology paired with AI creates another breakthrough category – by continuously analyzing smartwatch ECG data, photoplethysmography waveforms, and activity patterns, these systems can detect subclinical deterioration with 84.5% accuracy (AUC: 0.845), essentially serving as early warning systems for cardiac events^[98]. Perhaps most exciting is AI's emerging role in metabolic biomarker discovery – deep learning models are now identifying characteristic perturbations in fatty acid oxidation, glucose utilization, and mitochondrial function that predict both acute ischemic outcomes and long-term recovery trajectories^[99].

Technical innovations in AI-biomarker analysis

AI-Driven biomarker discovery: technical approaches, and innovations: The engine behind these advances is a suite of cutting-edge AI methodologies specifically adapted for cardiac applications. Multimodal architectures represent a quantum leap – these sophisticated neural networks can simultaneously process and correlate data from ECGs, cardiac MRI sequences, proteomic assays, and clinical variables to generate comprehensive risk assessments^[100,101]. Explainability has become equally crucial – techniques like SHAP (SHapley Additive exPlanations) analysis quantify exactly how much each biomarker contributes to predictions, whether it's the action potential duration (APD90) in drug safety screening or specific ST-segment features in ischemia detection^[21]. For chronic disease management, temporal modeling approaches now track biomarker evolution over time, identifying subtle patterns that foreshadow heart failure decompensation weeks before clinical deterioration becomes apparent^[102]. These technical innovations are transforming biomarkers from static measurements into dynamic, multidimensional indicators of cardiac health.

Enhancing clinical workflow and AI integration in biomarker analysis: AI technologies are not just revolutionizing biomarkers themselves but also improving their integration into clinical workflows. For example, point-of-care diagnostics have seen particularly rapid adoption – next-generation wearable patches and smartwatches now incorporate edge AI processors that analyze ECG rhythms in real-time, providing instant feedback about potential ischemic changes while patients go about their daily lives^[98]. In hospital settings, AI-powered metabolic phenotyping is personalizing post-AMI care – by analyzing patterns in a patient's cardiac substrate utilization (like preferential glucose vs fatty acid metabolism), these systems can predict recovery trajectories and guide tailored rehabilitation programs^[99]. Pharmaceutical development has also embraced AI biomarkers; the Comprehensive in vitro proarrhythmia Assay (CiPA) initiative now uses machine learning models that incorporate 11 key electrophysiological parameters (including dVm/dtmax and calcium handling metrics) to screen new drugs for arrhythmia risk with over 90% accuracy^[103].

Challenges and the future of AI in biomarkers

Despite remarkable progress, significant hurdles remain for widespread implementation. Validation represents the foremost

challenge – while single-center studies show impressive results (like the QCG™ system's performance), large multicenter trials with diverse patient populations are still needed to prove generalizability across different demographics and healthcare settings^[104]. Equity concerns also loom large – many AI models demonstrate reduced accuracy in underrepresented groups due to biases in training data, requiring intentional efforts to build more inclusive datasets^[102]. On the regulatory front, agencies are racing to establish appropriate frameworks – current guidance like the FDA's AI/ML Software as a Medical Device Action Plan provides a starting point but will need continuous updates to address emerging challenges in algorithm transparency, real-world monitoring, and post-market surveillance^[21]. These concerns must be addressed to facilitate the broader adoption of AI in clinical practice. The path forward will require unprecedented collaboration between clinicians, data scientists, regulators, and patients to ensure these powerful technologies fulfill their potential without compromising safety or equity. Looking ahead, AI's role in discovering novel biomarkers, integrating multimodal data, and personalizing treatment protocols will continue to evolve and refine the future of cardiac care.

Table 3 highlights AI's strong promise in biomarker-based diagnosis and risk prediction for ischemic heart disease. Large multicenter studies, such as CoDE-ACS^[105,106], demonstrated very high accuracy for MI diagnosis and rule-out (AUC ~ 0.95–0.97), with superior efficiency compared to guideline-based pathways. However, all remain retrospective and assay-specific. Earlier work also showed robust MI and 1-year event prediction, but highlighted calibration challenges and subgroup variability^[107]. Beyond ischemia, AI has extended into non-invasive biomarker surrogates: ECG-based prediction of natriuretic peptides^[108,109] and echo-based lab surrogates^[110] provide scalable prognostic tools, while novel devices such as spectrophotometric wearables^[111] or portable assays^[112] suggest new frontiers in rapid biomarker acquisition. Studies also explore broader CAD and HF risk stratification using clinical, biochemical, or cytokine data^[113–115], with generally encouraging but early results. Agreement across the literature lies in demonstrating that machine learning can extract richer diagnostic and prognostic value than conventional thresholds or scores. Where findings diverge is in scale and rigor: small, single-center proof-of-concepts often report near-perfect performance, while large multicenter validations show strong but more modest accuracy, and generalizability across assays, populations, and devices remains uncertain. Critical gaps persist regarding prospective deployment, calibration across diverse demographics, and demonstration of true outcome benefit; underscoring the need for multicenter impact trials, vendor-independent validation, and standardized reporting of fairness and performance metrics.

AI in risk prediction and assessment

AI-driven advances in risk assessment

Cardiovascular diseases (CVDs), including acute myocardial infarction (AMI) and heart failure (HF), remain the leading causes of global morbidity and mortality^[80,119,120]. While traditional risk scores like Framingham rely on limited clinical variables, artificial intelligence (AI) is revolutionizing risk prediction by integrating complex, multimodal data^[114,121,122].

Table 3

AI in biomarker analysis and risk prediction: current applications and research insights.

AI in biomarker analysis and risk prediction								
Study (author, year)	Population/dataset	Data modality	Purpose	AI method	Validation	Key performance	Key findings	Limitations
Doudesis <i>et al.</i> , 2023 (Nat Med) ^[109]	10 038 derivation; 10 286 across 7 cohorts	hs-cTnI + clinical variables	MI diagnosis & risk stratification	XGBoost (injury-stratified)	Multinational external	AUC: 0.95–0.97	Safely ruled out 61% with NPV 99.6%; consistent across systems	Abbott assay only; mostly White cohorts; retrospective
Boedinghaus <i>et al.</i> , 2024 ^[106]	4105 suspected NSTEMI (12 centers, 5 countries)	hs-cTnI 0/1/2 h + clinical variables	Early rule-out/rule-in T1MI	CoDE-ACS (4-model ensemble)	Multicenter retrospective	AUC: 0.95–0.97	Rule-out 56–68% vs ESC 25–35%; improved PPV for rule-in	Abbott assay only; retrospective care; no prospective trial
Doudesis <i>et al.</i> , 2022 ^[107]	20 761 suspected ACS (10 UK hospitals)	Serial hs-cTnI + age/sex	MI diagnosis + 1-year outcomes	Gradient boosting	Multicenter retrospective	AUC: 0.95	High-sensitivity rule-out; predicted 1-year M/CV death	Limited variables; poor intermediate calibration; no ethnicity data
Toprak <i>et al.</i> , 2024 ^[116]	2560 suspected MI (US + Australia)	POC hs-cTnI + 8 clinical variables	Rapid ED MI rule-out	Gradient boosting	Prospective cohorts, retrospective analysis	NPV 99.96%; Sens 99.7%	Outperformed ESC/ACC 0 hour pathways; effective in early presenters	Siemens POC only; not prospectively deployed; missing ethnicity data
Han <i>et al.</i> , 2024 ^[112]	72 serum samples (blind val $n = 66$)	Paper-based assay images	Ultra-sensitive cTnI detection	4-level FC neural nets	Small blind validation	LoD 0.16 pg/mL; $r = 0.984$ vs analyzer	Portable (low-cost ~\$4.25/test); 25-min workflow	Small sample; slow vs rapid LFAs; low-level misclassification
Sengupta <i>et al.</i> , 2023 ^[111]	238 ACS pts (5 Indian sites)	Wrist-worn spectrophotometry	Predict elevated hs-cTnI non-invasively	Hybrid CNN–RNN	Small multicenter	AUC: 0.90–0.92	Bloodless rapid test feasible; associated with CAD and WMAs	Small sample; motion/skin confounders; no serial patterns
Kucukakcali <i>et al.</i> , 2025 ^[117]	1319 Iraqi pts	8 clinical/biochemical vars	Interpretable MI diagnosis	Explainable Boosting Machine	Single-center	AUC: 0.98; Acc 97%	Fully interpretable; troponin/CK-MB major drivers	Local dataset only; no external validation
Raparelli <i>et al.</i> , 2023 ^[114]	311 angiography pts (Italy)	Clinical + psychosocial + cytokines	Identify obstructive vs non-obstructive CAD	XGBoost + SHAP	Single-center	Acc 83%	Showed sex-specific inflammatory predictors	Small cohort; selection bias; no ischemia testing
Johri <i>et al.</i> , 2021 ^[115]	459 angiography pts	Carotid US + clinical vars	Predict CAD + 30-day events	RF ^[118] + RSF (events)	Single-center	CAD AUC: 0.97; Events C-index 0.86	Improved event prediction vs Cox	High-risk cohort; short follow-up
Nayazi <i>et al.</i> , 2024 ^[108]	8253 dev; 6821 external	12-lead ECG	Predict NT-proBNP & CVD risk	1D ResNet	Large multi-cohort	$r = 0.57\text{--}0.66$; HF AUROC 0.79–0.82	ECG-based pNT-proBNP comparable to measured NT-proBNP	European-only; black-box model; assay differences
Liu <i>et al.</i> , 2023 ^[109]	20 789 + 5032 dev; 4001 int; 6042 ext	12-lead ECG	Predict BNP/NT-proBNP & mortality	ECG12Net CNN	Internal + external	AUC: 0.85–0.89	ECG-pBNP correlated well; predicted mortality	Taiwan-only; no echo correlation; assay-specific
Hughes <i>et al.</i> , 2021 ^[110]	70 066 echo videos; external 1301	2D A4C echo	Predict BNP, troponin, Hb, BUN	2 + 1D ResNet-34	Large internal + external	BNP AUC: 0.86	Echo-estimated BNP predicted HF hospitalization	Echo-only view; timing mismatch; retrospective
Sun <i>et al.</i> , 2024 ^[113]	200 CHF pts	65 clinical/lab/echo vars	Predict 3-mo worsening HF	Random Forest	Single-center	AUC: 0.84	5-variable risk tool developed; strong sensitivity	Small sample; risk of overfitting; no external validation

2D/(2 + 1D): two-dimensional/two-plus-one-dimensional imaging; ACS: acute coronary syndrome; AF: atrial fibrillation; AI: artificial intelligence; AUC: area under the receiver operating characteristic curve; BNP: B-type natriuretic peptide; BP: blood pressure; BUN: blood urea nitrogen; CEUS: contrast-enhanced ultrasound; CHF: chronic heart failure; C-index: concordance index; CK-MB: creatine kinase-MB; CNN: convolutional neural network; CODE-ACS: code for diagnosis and evaluation in acute coronary syndrome; Cr: creatinine; CVD: cardiovascular disease; ECG: electrocardiography; ED: emergency department; ESC: European Society of Cardiology; GAML: Generalized Additive Model; Hb: hemoglobin; HCHS: Hamburg City Health Study; HF: heart failure; High-STEACS: high-sensitivity troponin in evaluation of acute coronary syndrome; HR: heart rate; hs-cTnI: high-sensitivity cardiac troponin I; IPN: Intrima-Media Plaque Number for index; context-specific; LoD: limit of detection; MI: myocardial infarction; ML: machine learning; NPV: negative predictive value; NT-proBNP/pBNP: N-terminal pro-B-type natriuretic peptide; OR: odds ratio; POC: Point-of-Care; PPV: positive predictive value; QoL: quality of life; r: Pearson correlation coefficient; ResNet-34: 34-layer residual neural network; RF: random forest; RNN: recurrent neural network; RSF: random survival forest; SE: squeeze-and-excitation (neural network block); SHAP: SHapley Additive exPlanations; SHIP-START: Study of Health in Pomerania START cohort; SHIP-TREND: Study of Health in Pomerania TREND cohort; UA: uric acid; WHF: worsening heart failure; XGBoost: extreme gradient boosting.

By analyzing everything from coronary imaging to gut microbiome profiles, AI models can identify high-risk patients earlier and with greater precision than ever before^[19,115,123]. However, significant challenges in validation and implementation must be addressed before these tools can reach their full potential in clinical practice^[80,120].

The true power of AI lies in its ability to synthesize diverse data types that were previously analyzed in isolation^[119,124]. Imaging-based AI has shown remarkable accuracy, with carotid ultrasound algorithms detecting intraplaque neovascularization (a marker of vulnerable plaques) at 97% accuracy; far surpassing human interpretation^[115]. Similarly, AI applied to coronary CT angiography can now predict 30-day cardiovascular events better than traditional risk scores, particularly in elderly patients with multiple comorbidities^[19,125]. Beyond imaging, wearable devices equipped with machine learning can detect subtle ECG changes predictive of arrhythmias or ischemia, often days before symptoms appear^[119,126]. Perhaps most promising is AI's ability to integrate emerging biomarkers like inflammatory cytokines (IL-1 β , IL-33) and gut microbiome patterns with traditional risk factors, enabling truly personalized risk assessments^[114,121,122]. For diabetic patients at cardiovascular risk, these models can now predict microvascular complications with 91% accuracy by analyzing continuous glucose monitoring data alongside retinal scans and ECG findings^[123].

Clinical integration of AI-based risk prediction

Several machine-learning approaches have demonstrated particular success in cardiovascular applications^[114,115,126]. The XGBoost algorithm, when combined with SHAP explainability tools, achieves 83% accuracy in distinguishing between obstructive and non-obstructive coronary artery disease by weighing clinical, biochemical, and psychosocial factors^[114]. For predicting short-term events, Random Survival Forest models have shown a 17.8% improvement over conventional statistical methods in Medicare populations^[115,125]. Deep learning has transformed ECG interpretation, with convolutional neural networks now detecting heart failure from raw ECG traces as accurately as cardiologists^[126]. However, the field faces a validation crisis – a comprehensive review of 486 AI models found that only 10 met rigorous independent validation standards^[127]. Many models perform well in controlled trials but falter in real-world settings due to inadequate diversity in training data (most come from North America and Europe) or failure to account for evolving patient phenotypes over time^[80,127].

Future directions in AI-based risk prediction

The next generation of AI tools must prioritize clinical utility and equity^[80,120,124]. Emerging solutions include federated learning systems that allow hospitals to collaboratively improve models without sharing sensitive patient data, and “living” algorithms that continuously adapt to new information^[124]. Regulatory bodies are now developing certification frameworks to evaluate AI tools for bias and generalizability before clinical deployment^[80]. Looking ahead, the integration of genomic data with real-time wearable monitoring could enable dynamic risk scores that update hourly, while quantum computing may soon unlock the ability to model previously intractable interactions between thousands of variables^[121,124]. For AI to fulfill its

promise, the cardiovascular community must address critical challenges, including reimbursement policies, clinician trust in “black box” algorithms, and the ethical implications of predictive medicine^[80,120]. When these hurdles are overcome, AI-powered risk prediction will fundamentally transform cardiovascular care, shifting from reactive treatment to truly personalized prevention^[120,124].

Figure 1 shows how AI enhances ischemic heart disease diagnosis across ECG, imaging, biomarkers, and risk assessment, reflecting its growing role in clinical practice.

Discussion

Across ECG, imaging, and biomarker-based applications, most current AI tools for ischemic heart disease are supported primarily by retrospective evidence, with only a limited number of small prospective or real-world studies^[128]. Retrospective multi-center models, including deep-learning ECG classifiers, CCTA radiomics, and troponin-augmented machine-learning scores, consistently demonstrate strong discriminative performance and external validation^[129–131]. However, the generalizability of many models decreases when applied outside controlled development environments, in part due to demographic and geographic homogeneity in training datasets and the lack of standardized validation frameworks^[38,132,133]. Clinical integration remains a major challenge. Although several cardiovascular AI solutions – such as automated EF estimation, commercial arrhythmia detectors, and FFR-CT – have obtained regulatory approval, most AI systems targeting ischemic heart disease have not yet achieved FDA or EMA clearance^[134,135]. Regulatory pathways increasingly emphasize dataset diversity, calibration stability, explainability, and post-deployment monitoring, all of which are insufficiently addressed in many current studies^[136,137]. Real-world implementation also requires seamless workflow integration, transparent reporting, and robust governance around data quality, privacy, and algorithmic fairness^[138,139]. Taken together, these observations highlight the need for large-scale prospective trials, regulatory-grade validation, and systematic post-market surveillance to ensure clinical safety and effectiveness. As the field evolves, AI has strong potential to improve early diagnosis, triage, and longitudinal risk assessment – provided that future models prioritize reproducibility, inclusiveness, and clear demonstration of clinical utility.

This review offers a comprehensive, cross-modality overview of AI in ischemic heart disease and, through a structured search with standardized tables, provides a clear and coherent synthesis across ECG, imaging, biomarker, and risk-prediction domains. Despite these strengths, several limitations should be acknowledged. This work is a structured narrative review rather than a full systematic review, and some relevant studies may not have been captured. The included literature is heterogeneous in design, datasets, and validation rigor, precluding quantitative pooling and limiting direct comparisons across studies. Most available evidence remains retrospective, often with incomplete reporting or limited external validation, which constrains conclusions about real-world performance. Additionally, AI technologies evolve rapidly, and newer tools or changing regulatory statuses may have emerged after the search period.

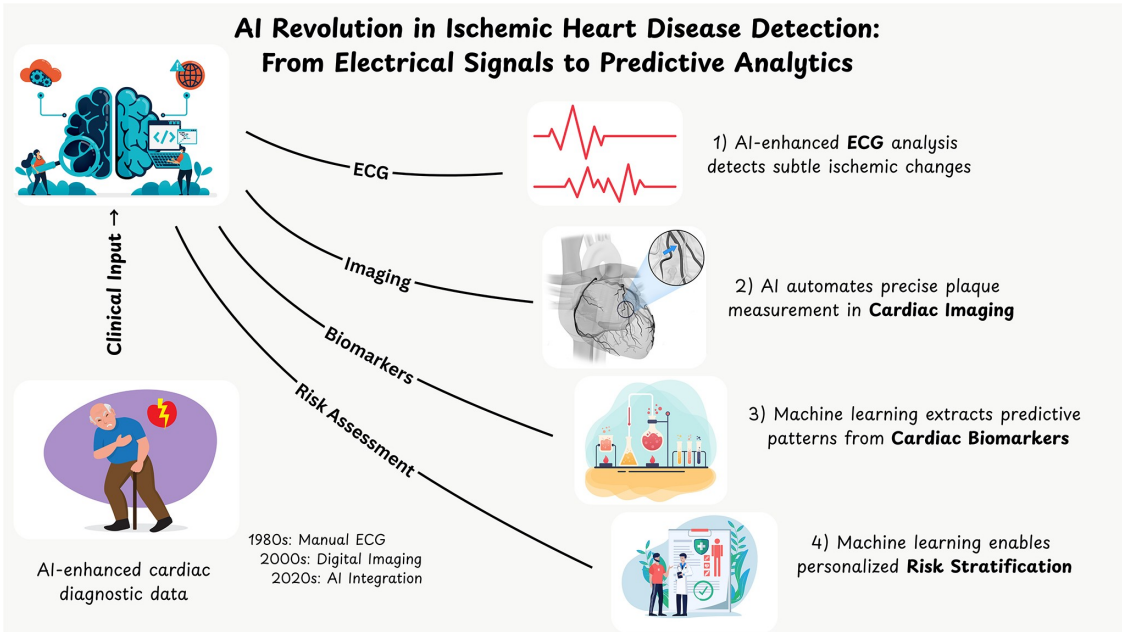


Figure 1. Artificial intelligence in ischemic heart disease diagnosis across eras and modalities. The figure illustrates how AI enhances clinical decision-making through improved analysis of ECG signals, cardiac imaging, biomarkers, and risk assessment, marking a shift from traditional methods to integrated predictive analytics. Five images are from Vecteezy.Com (Vectors by Vecteezy, free license), and one image is from Wikimedia Commons (by BruceBlaus, CC BY-SA 4.0).

Conclusion

AI applications across ECG analysis, cardiac imaging, and bio-marker-based risk prediction show strong potential for improving early detection and stratification of ischemic heart disease. Many models demonstrate excellent retrospective performance and external validation; however, prospective evidence remains limited, and real-world clinical impact has yet to be fully established. To bridge this gap, future work should prioritize large-scale prospective trials, standardized reporting, and transparent validation frameworks aligned with regulatory expectations. With rigorous evaluation and thoughtful clinical integration, AI has the potential to become a powerful tool for timely diagnosis, risk assessment, and personalized management of ischemic heart disease.

Ethical approval

As this study is a narrative review, ethical approval was not required.

Consent

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Author contributions

P.A.: writing – original draft, writing – review and editing, conceptualization, data curation, supervision; J.K.: writing – original draft, writing – review and editing, conceptualization, data curation; I.R.-J.: writing – original draft, writing – review and editing, data curation.

Conflicts of interest disclosure

The authors declare no conflicts of interest.

Research registration unique identifying number (UIN)

This is a narrative review and does not involve human subjects or patient data; therefore, registration was not required.

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All data are publicly available.

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References

- [1] Mechanic OJ, Gavin M, Grossman SA. Acute myocardial infarction. *J Emerg Med* 2017;53:391–96.
- [2] Bajaj A, Sethi A, Rathor P, *et al.* Acute complications of myocardial infarction in the current era: diagnosis and management. *J Invest Med* 2015;63:844–55.
- [3] Wassie M, Lee M-S, Sun BC, *et al.* Single vs serial measurements of cardiac troponin level in the evaluation of patients in the emergency department with suspected acute myocardial infarction. *JAMA Network Open* 2021;4:e2037930e.
- [4] Berte B, Duytschaever M, Elices J, *et al.* Variability in interpretation of the electrocardiogram in young athletes: an unrecognized obstacle for electrocardiogram-based screening protocols. *Europace* 2015;17:1435–40.
- [5] Omerovic S, Jain A. Echocardiogram. In: StatPearls [Internet]. StatPearls Publishing; 2023.
- [6] Olawade DB, Aderinto N, Olatunji G, *et al.* Advancements and applications of Artificial Intelligence in cardiology: current trends and future prospects. *J Med Surg Public Health* 2024;3:100109.
- [7] Ali MT, Ali U, Ali S, *et al.* Transforming cardiac care: AI and machine learning innovations. *Int J Multidiscip Res Growth Eval* 2024;5:379–84.
- [8] Najjar R. Redefining radiology: a review of Artificial Intelligence integration in medical imaging. *Diagnostics* 2023;13:2760.
- [9] Uday IA, Hassan O. AI-driven technology in heart failure detection and diagnosis: a review of the advancement in personalized healthcare. *Symmetry* 2025;17:469.
- [10] Sumalatha U, Prakasha KK, Prabhu S, *et al.* Deep learning applications in ECG analysis and disease detection: an investigation study of recent advances. *IEEE Access* 2024;12:126258–84.
- [11] Khalifa M, Albadawy M. AI in diagnostic imaging: revolutionising accuracy and efficiency. *Comp Method Programs Biomed Update* 2024;5:100146.
- [12] Bhandari A. Revolutionizing Radiology With Artificial Intelligence. *Cureus* 2024;16:e72646.
- [13] Schulze K, Stantien A-M, Williams MC, *et al.* Coronary CT angiography evaluation with artificial intelligence for individualized medical treatment of atherosclerosis: a consensus statement from the QCI study group. *Nat Rev Cardiol* 2025;22:1–16.
- [14] Pasdeloup D, Østvik A, Olaisen S, *et al.* Challenges and strategies for deep learning in cardiovascular imaging: ejection fraction and heart failure management. *Cardiovasc Imaging* 2025;18:751–64.
- [15] Raissi-Dehkordi N, Raissi-Dehkordi N, Xu B. Contemporary applications of artificial intelligence and machine learning in echocardiography. *Npj Cardiovasc Health* 2025;2:1–10.
- [16] Aromiwura AA, Settle T, Umer M, *et al.* Artificial intelligence in cardiac computed tomography. *Progress in cardiovascular diseases. Prog Cardiovasc Dis* 2023;81:54–77.
- [17] Hu X, Zhang H, Caobelli F, *et al.* The role of deep learning in myocardial perfusion imaging for diagnosis and prognosis: a systematic review. *iScience* 2024;27:111374.
- [18] Lin A, Manral N, McElhinney P, *et al.* Deep learning-enabled coronary CT angiography for plaque and stenosis quantification and cardiac risk prediction: an international multicentre study. *The Lancet Digital Health* 2022;4:e256–e65.
- [19] Brandt V, Emrich T, Schoepf UJ, *et al.* Ischemia and outcome prediction by cardiac CT based machine learning. *Int J Cardiovasc Imaging* 2020;36:2429–39.
- [20] Elias P, Jain SS, Poterucha T, *et al.* Artificial intelligence for cardiovascular care – part 1: advances: JACC review topic of the week. *J Am Coll Cardiol* 2024;83:2472–86.
- [21] Jain SS, Elias P, Poterucha T, *et al.* Artificial intelligence in cardiovascular care – part 2: applications: JACC review topic of the week. *J Am Coll Cardiol* 2024;83:2487–96.
- [22] Tilea I, Varga A, Serban RC. Past, present, and future of blood biomarkers for the diagnosis of acute myocardial infarction – promises and challenges. *Diagnostics* 2021;11:881.
- [23] Kim SY, Lee JP, Shin WR, *et al.* Cardiac biomarkers and detection methods for myocardial infarction. *Mol Cell Toxicol* 2022;18:443–55.
- [24] Kim J, Shon B, Kim S, *et al.* ECG data analysis to determine ST-segment elevation myocardial infarction and infarction territory type: an integrative approach of artificial intelligence and clinical guidelines. *Front Physiol* 2024;15:1462847.
- [25] Cook DA, Oh SY, Pusic MV. Accuracy of physicians' electrocardiogram interpretations: a systematic review and meta-analysis. *JAMA Intern Med* 2020;180:1461–71.
- [26] Iskander J, Kelada P, Rashad L, *et al.* Advanced echocardiography techniques: the future stethoscope of systemic diseases. *Curr Probl Cardiol* 2022;47:100847.
- [27] Huson MAM, Kaminstein D, Kahn D, *et al.* Cardiac ultrasound in resource-limited settings (CURLS): towards a wider use of basic echo applications in Africa. *Ultrasound J* 2019;11:34.
- [28] Pourali A, Omid Y. Diagnosis of acute myocardial infarction: highlighting cardiac troponins as vital biomarkers. *Bioimpacts* 2023;13:85–88.
- [29] Daubert MA, Jeremias A. The utility of troponin measurement to detect myocardial infarction: review of the current findings. *Vasc Health Risk Manag* 2010;6:691–99.
- [30] Yang X, Li J, Hu D, *et al.* Predicting the 10-year risks of atherosclerotic cardiovascular disease in Chinese population: the China-PAR project (prediction for ASCVD risk in China). *Circulation* 2016;134:1430–40.
- [31] Kasartzian D-I, Tsiampalis T. Transforming cardiovascular risk prediction: a review of machine learning and artificial intelligence innovations. *Life* 2025;15:94.
- [32] Ekundayo F, Nyavor H. AI-driven predictive analytics in cardiovascular diseases: integrating big data and machine learning for early diagnosis and risk prediction. *Int J Res Publ Rev* 2024;5:1240–56.
- [33] Attia ZI, Kapa S, Lopez-Jimenez F, *et al.* Screening for cardiac contractile dysfunction using an artificial intelligence-enabled electrocardiogram. *Nat Med* 2019;25:70–74.
- [34] Knoery CR, Bond R, Iftikhar A, *et al.* SPICED-ACS: study of the potential impact of a computer-generated ECG diagnostic algorithmic certainty index in STEMI diagnosis: towards transparent AI. *J Electrocardiol* 2019;57:s86–s91.
- [35] Wu YH, Li AH, Chen TC, *et al.* Compared with physician overread, computer is less accurate but helpful in interpretation of electrocardiography for ST-segment elevation myocardial infarction. *J Electrocardiol* 2023;81:60–65.
- [36] Makimoto H, Höckmann M, Lin T, *et al.* Performance of a convolutional neural network derived from an ECG database in recognizing myocardial infarction. *Sci Rep* 2020;10:8445.
- [37] Cao Y, Liu W, Zhang S, *et al.* Detection and localization of myocardial infarction based on multi-scale ResNet and attention mechanism. *Front Physiol* 2022;13:783184.
- [38] Ribeiro AH, Ribeiro MH, Paixão GMM, *et al.* Automatic diagnosis of the 12-lead ECG using a deep neural network. *Nat Commun* 2020;11:1760.
- [39] Wang J, Guo X. Automated detection of myocardial infarction based on an improved state refinement module for LSTM/GRU. *Artif Intell Med* 2024;152:102865.
- [40] US Food and Drug Administration. Artificial Intelligence and Machine Learning in Software as a Medical Device. US Food & Drug Administration, Silver Spring; 2021.
- [41] Ayyad M, Albandak M, Gala D, *et al.* Reevaluating STEMI: the utility of the occlusive myocardial infarction classification to enhance management of acute coronary syndromes. *Curr Cardiol Rep* 2025;27:75.
- [42] Baker PO, Karim SR, Smith SW, *et al.* Artificial Intelligence driven prehospital ECG interpretation for the reduction of false positive emergent cardiac catheterization lab activations: a retrospective cohort study. *Prehosp Emerg Care* 2025;29:218–26.
- [43] Goebel M, Vaida F, Kahn C, *et al.* A novel algorithm for improving the diagnostic accuracy of prehospital ST-elevation myocardial infarction. *Prehosp Disaster Med* 2019;34:489–96.
- [44] Goebel M, Westafer LM, Ayala SA, *et al.* A novel algorithm for improving the prehospital diagnostic accuracy of ST-segment elevation myocardial infarction. *Prehosp Disaster Med* 2024;39:37–44.
- [45] Han C, Song Y, Lim HS, *et al.* Automated detection of acute myocardial infarction using asynchronous electrocardiogram signals-preview of implementing Artificial Intelligence with multichannel electrocardiographs obtained from smartwatches: retrospective study. *J Med Internet Res* 2021;23:e31129.
- [46] Zeng W, Yuan J, Yuan C, *et al.* Classification of myocardial infarction based on hybrid feature extraction and artificial intelligence tools by adopting tunable-Q wavelet transform (TQWT), variational mode

- decomposition (VMD) and neural networks. *Artif Intell Med* 2020;106:101848.
- [47] Poterucha TJ, Jing L, Ricart RP, *et al.* Detecting structural heart disease from electrocardiograms using AI. *Nature* 2025;644:221–30.
- [48] Sau A, Pastika L, Sieliwonczyk E, *et al.* Artificial intelligence-enabled electrocardiogram for mortality and cardiovascular risk estimation: a model development and validation study. *The Lancet Digital Health* 2024;6:e791–e802.
- [49] Fiorina L, Carbonati T, Narayanan K, *et al.* Near-term prediction of sustained ventricular arrhythmias applying artificial intelligence to single-lead ambulatory electrocardiogram. *Eur Heart J* 2025;46:1998–2008.
- [50] Jain SS, Sun T, Brown K, *et al.* Antiracist ai: a machine learning model using electrocardiograms and echocardiograms can detect transthyretin cardiac amyloidosis and decrease racial bias in diagnostic testing. *J Am Coll Cardiol* 2023;81:338–.
- [51] Zeng W, Yuan C. Myocardial infarction detection using ITD, DWT and deterministic learning based on ECG signals. *Cogn Neurodyn* 2023;17:941–64.
- [52] Awasthi S, Sachdeva N, Gupta Y, *et al.* Identification and risk stratification of coronary disease by artificial intelligence-enabled ECG. *EClinicalMedicine* 2023;65:102259.
- [53] Ose B, Sattar Z, Gupta A, *et al.* Artificial Intelligence interpretation of the electrocardiogram: a state-of-the-art review. *Curr Cardiol Rep* 2024;26:561–80.
- [54] Siontis KC, Noseworthy PA, Attia ZI, *et al.* Artificial intelligence-enhanced electrocardiography in cardiovascular disease management. *Nat Rev Cardiol* 2021;18:465–78.
- [55] He Y, Zhou Y, Qian Y, *et al.* Cardioattentionnet: advancing ECG beat characterization with a high-accuracy and portable deep learning model. *Front Cardiovasc Med* 2025;11:1473482.
- [56] Zhang J, Gajjala S, Agrawal P, *et al.* Fully automated echocardiogram interpretation in clinical practice. *Circulation* 2018;138:1623–35.
- [57] Schindler TH, Fearon WF, Pelletier-Galarneau M, *et al.* Myocardial perfusion PET for the detection and reporting of coronary microvascular dysfunction: a JACC: cardiovascular imaging expert panel statement. *JACC Cardiovasc Imaging* 2023;16:536–48.
- [58] Topol EJ. High-performance medicine: the convergence of human and artificial intelligence. *Nat Med* 2019;25:44–56.
- [59] Kuwahara A, Iwasaki Y, Kobayashi M, *et al.* Artificial intelligence-derived left ventricular strain in echocardiography in patients treated with chemotherapy. *Int J Cardiovasc Imaging* 2024;40:1903–10.
- [60] Stewart JE, Goudie A, Mukherjee A, *et al.* Artificial intelligence-enhanced echocardiography in the emergency department. *Emerg Med Australas* 2021;33:1117–20.
- [61] Seifert R, Weber M, Kocakavuk E, *et al.* editors. Artificial Intelligence and Machine Learning in Nuclear Medicine: Future Perspectives. *Seminars in Nuclear Medicine*. Elsevier; 2021.
- [62] Joshi M, Melo DP, Ouyang D, *et al.* Current and future applications of artificial intelligence in cardiac CT. *Curr Cardiol Rep* 2023;25:109–17.
- [63] Jiang B, Guo N, Ge Y, *et al.* Development and application of artificial intelligence in cardiac imaging. *Br J Radiol* 2020;93:20190812.
- [64] Nørgaard Bjarne L, Leipsic J, Gaur S, *et al.* Diagnostic performance of noninvasive fractional flow reserve derived from coronary computed tomography angiography in suspected coronary artery disease. *JACC* 2014;63:1145–55.
- [65] Yang J, Shan D, Wang X, *et al.* On-site computed tomography-derived fractional flow reserve to guide management of patients with stable coronary artery disease: the TARGET randomized trial. *Circulation* 2023;147:1369–81.
- [66] Farhad A, Reza R, Azamossadat H, *et al.* Artificial intelligence in estimating fractional flow reserve: a systematic literature review of techniques. *BMC Cardiovasc Disord* 2023;23:407.
- [67] Peters B, Paul JF, Symons R, *et al.* Invasive fractional-flow-reserve prediction by coronary CT angiography using artificial intelligence vs computational fluid dynamics software in intermediate-grade stenosis. *Int J Cardiovasc Imaging* 2024;40:1875–80.
- [68] Van den Oever L, Vonder M, Van Assen M, *et al.* Application of artificial intelligence in cardiac CT: from basics to clinical practice. *Eur J Radiol* 2020;128:108969.
- [69] Cau R, Cherchi V, Micheletti G, *et al.* Potential role of artificial intelligence in cardiac magnetic resonance imaging: can it help clinicians in making a diagnosis? *J Thorac Imaging* 2021;36:142–48.
- [70] Hernandez KAL, Rienmüller T, Baumgartner D, *et al.* Deep learning in spatiotemporal cardiac imaging: a review of methodologies and clinical usability. *Comput Biol Med* 2021;130:104200.
- [71] Oikonomou EK, Williams MC, Kotanidis CP, *et al.* A novel machine learning-derived radiotranscriptomic signature of perivascular fat improves cardiac risk prediction using coronary CT angiography. *Eur Heart J* 2019;40:3529–43.
- [72] Miller RJ, Huang C, Liang JX, *et al.* Artificial intelligence for disease diagnosis and risk prediction in nuclear cardiology. *J Nucl Cardiol* 2022;29:1754–62.
- [73] Krajcer Z. Artificial intelligence in cardiovascular medicine: historical overview, current status, and future directions. *Tex Heart Inst J* 2022;49:e207527.
- [74] Baeßler B, Engelhardt S, Hekalo A, *et al.* Perfect match: radiomics and artificial intelligence in cardiac imaging. *Circulation* 2024;17:e015490.
- [75] Mannil M, Eberhard M, Von Spiczak J, *et al.* Artificial intelligence and texture analysis in cardiac imaging. *Curr Cardiol Rep* 2020;22:1–9.
- [76] Litjens G, Kooi T, Bejnordi BE, *et al.* A survey on deep learning in medical image analysis. *Med Image Anal* 2017;42:60–88.
- [77] Salih A, Boscolo Galazzo I, Gkontra P, *et al.* Explainable artificial intelligence and cardiac imaging: toward more interpretable models. *Circulation* 2023;16:e014519.
- [78] Sengupta PP, Dey D, Davies RH, *et al.* Challenges for augmenting intelligence in cardiac imaging. *The Lancet Digital Health* 2024;6:e739–e48.
- [79] Yu AC, Mohajer B, Eng J. External validation of deep learning algorithms for radiologic diagnosis: a systematic review. *Radiology* 2022;4:e210064.
- [80] Cai Y, Cai Y-Q, Tang L-Y, *et al.* Artificial intelligence in the risk prediction models of cardiovascular disease and development of an independent validation screening tool: a systematic review. *BMC Med* 2024;22:56.
- [81] Kusunose K. Steps to use artificial intelligence in echocardiography. *J Echocardiogr* 2021;19:21–27.
- [82] Giannopoulos AA, Keller L, Sepulcri D, *et al.* High-speed on-site deep learning-based FFR-CT algorithm: evaluation using invasive angiography as the reference standard. *Am J Roentgenol* 2023;221:460–70.
- [83] Qi H, Qian P, Tang L, *et al.* Predicting late gadolinium enhancement of acute myocardial infarction in contrast-free cardiac cine MRI using deep generative learning. *Circulation* 2024;17:e016786.
- [84] Amini M, Pursamimi M, Hajianfar G, *et al.* Machine learning-based diagnosis and risk classification of coronary artery disease using myocardial perfusion imaging SPECT: a radiomics study. *Sci Rep* 2023;13:14920.
- [85] Liu X, Fan Y, Li S, *et al.* Deep learning-based automated left ventricular ejection fraction assessment using 2-D echocardiography. *Am J Physiol Heart Circ Physiol* 2021;321:H390–H9.
- [86] Vega R, Kwok C, Rakkunedeth Hareendranathan A, *et al.* Assessment of an artificial intelligence tool for estimating left ventricular ejection fraction in echocardiograms from apical and parasternal long-axis views. *Diagnostics* 2024;14:1719.
- [87] Xu W, Shi J, Lin Y, *et al.* Deep learning-based image segmentation model using an MRI-based convolutional neural network for physiological evaluation of the heart. *Front Physiol* 2023;14:1148717.
- [88] Ben Khalifa A, Mili M, Maatouk M, *et al.* Deep transfer learning for classification of late gadolinium enhancement cardiac MRI images into myocardial infarction, myocarditis, and healthy classes: comparison with subjective visual evaluation. *Diagnostics* 2025;15:207.
- [89] Bhav S, Rodriguez V, Poterucha T, *et al.* Deep learning to detect left ventricular structural abnormalities in chest X-rays. *Eur Heart J* 2024;45:2002–12.
- [90] Kim H, Yang YJ, Han K, *et al.* Validation of a deep learning-based software for automated analysis of T2 mapping in cardiac magnetic resonance imaging. *Quant Imaging Med Surg* 2023;13:6750.
- [91] Kalapos A, Szabó L, Dohy Z, *et al.* Automated T1 and T2 mapping segmentation on cardiovascular magnetic resonance imaging using deep learning. *Front Cardiovasc Med* 2023;10:1147581.
- [92] Lee C-K, Chen T-L, Wu J-E, *et al.* Multimodal deep learning models utilizing chest X-ray and electronic health record data for predictive screening of acute heart failure in emergency department. *Comput Methods Programs Biomed* 2024;255:108357.
- [93] Dey D, Slomka PJ, Leeson P, *et al.* Artificial Intelligence in cardiovascular imaging: JACC state-of-the-art review. *J Am Coll Cardiol* 2019;73:1317–35.

- [94] Baglioni V, Bozza F, Beatrice A, *et al.* Non-pharmacological treatments in paediatric migraine. *J Clin Med* 2024;13:1278.
- [95] Mahajan S, Dave H, Bothe S, *et al.* Objective monitoring of cardiovascular biomarkers using Artificial Intelligence (AI). *Asian J Pharm Res* 2022;12:229–34.
- [96] Ortiz DA, Loeffelholz MJ. Practical challenges of point-of-care testing. *Clin Lab Med* 2023;43:155–65.
- [97] De Santis MC, Bockorny B, Hirsch E, *et al.* Exploiting pancreatic cancer metabolism: challenges and opportunities. *Trends Mol Med* 2024;30:592–604.
- [98] Haghighyegh F, Norouziyazad A, Haghani E, *et al.* Revolutionary point-of-care wearable diagnostics for early disease detection and biomarker discovery through intelligent technologies. *Adv Sci (Weinh)* 2024;11:e2400595.
- [99] Chen A-T, Zhang Y, Zhang J. Artificial intelligence in cardiac metabolism: the next frontier in cardiovascular health. *Metabolism Target Organ Damage* 2025;5:N/A–N/A.
- [100] Chan Y-H, Zhao JV, Au Yeung S-L, *et al.* Early renal dysfunction in middle-aged adults predicts fatal and nonfatal cardiovascular events. *J Am Coll Cardiol* 2024;83:2711–13.
- [101] Khan MS, Arshad MS, Greene SJ, *et al.* Artificial intelligence and heart failure: a state-of-the-art review. *Eur J Heart Fail* 2023;25:1507–25.
- [102] Khera R, Oikonomou EK, Nadkarni GN, *et al.* Transforming cardiovascular care with artificial intelligence: from discovery to practice: JACC state-of-the-art review. *J Am Coll Cardiol* 2024;84:97–114.
- [103] Seliverstov PV, Kutsenko VP, Gorelova VG, *et al.* Using artificial intelligence for biomarker analysis in clinical diagnostics. *Molekulyarnaya Meditsina (Molecular Medicine)* 2024;22:31–39.
- [104] Park MJ, Choi YJ, Shim M, *et al.* Performance of ECG-derived digital biomarker for screening coronary occlusion in resuscitated out-of-hospital cardiac arrest patients: a comparative study between Artificial Intelligence and a group of experts. *J Clin Med* 2024;13:1354.
- [105] Doudehis D, Lee KK, Boeddinghaus J, *et al.* Machine learning for diagnosis of myocardial infarction using cardiac troponin concentrations. *Nat Med* 2023;29:1201–10.
- [106] Boeddinghaus J, Doudehis D, Lopez-Ayala P, *et al.* Machine learning for myocardial infarction compared with guideline-recommended diagnostic pathways. *Circulation* 2024;149:1090–101.
- [107] Doudehis D, Lee KK, Yang J, *et al.* Validation of the myocardial-ischaemic-injury-index machine learning algorithm to guide the diagnosis of myocardial infarction in a heterogenous population: a prespecified exploratory analysis. *The Lancet Digital Health* 2022;4:e300–e8.
- [108] Neyazi M, Bremer JP, Knorr MS, *et al.* Deep learning-based NT-proBNP prediction from the ECG for risk assessment in the community. *Clin Chem Lab Med (CCLM)* 2024;62:740–52.
- [109] Liu P-Y, Lin C, Lin C-S, *et al.* Artificial intelligence-enabled electrocardiography detects b-type natriuretic peptide and n-terminal pro-brain natriuretic peptide. *Diagnostics* 2023;13:2723.
- [110] Hughes JW, Yuan N, He B, *et al.* Deep learning evaluation of biomarkers from echocardiogram videos. *EBioMedicine* 2021;73:103613.
- [111] Sengupta S, Biswal S, Titus J, *et al.* A novel breakthrough in wrist-worn transdermal troponin-I-sensor assessment for acute myocardial infarction. *Eur Heart J-Digital Health* 2023;4:145–54.
- [112] Han GR, Goncharov A, Eryilmaz M, *et al.* Deep learning-enhanced chemiluminescence vertical flow assay for high-sensitivity cardiac troponin I testing. *Small* 2025;21:2411585.
- [113] Sun Z, Wang Z, Yun Z, *et al.* Machine learning-based model for worsening heart failure risk in Chinese chronic heart failure patients. *ESC Heart Fail* 2025;12:211–28.
- [114] Raparelli V, Romiti GF, Di Teodoro G, *et al.* A machine-learning based bio-psycho-social model for the prediction of non-obstructive and obstructive coronary artery disease. *Clin Res Cardiol* 2023;112:1263–77.
- [115] Johri AM, Mantella LE, Jamthikar AD, *et al.* Role of artificial intelligence in cardiovascular risk prediction and outcomes: comparison of machine-learning and conventional statistical approaches for the analysis of carotid ultrasound features and intra-plaque neovascularization. *Int J Cardiovasc Imaging* 2021;37:3145–56.
- [116] Toprak B, Solleder H, Di Carluccio E, *et al.* Diagnostic accuracy of a machine learning algorithm using point-of-care high-sensitivity cardiac troponin I for rapid rule-out of myocardial infarction: a retrospective study. *The Lancet Digital Health* 2024;6:e729–e38.
- [117] Kucukakcali Z, Cicek IB, Akbulut S. Interpretable prediction of myocardial infarction using explainable boosting machines: a biomarker-based machine learning approach. *Diagnostics* 2025;15:2219.
- [118] Lin A, van Diemen PA, Motwani M, *et al.* Machine learning from quantitative coronary computed tomography angiography predicts fractional flow reserve-defined ischemia and impaired myocardial blood flow. *Circulation* 2022;15:e014369.
- [119] Huang JD, Wang J, Ramsey E, *et al.* Applying artificial intelligence to wearable sensor data to diagnose and predict cardiovascular disease: a review. *Sensors* 2022;22:8002.
- [120] Lüscher TF, Wenzl FA, D'Ascenzo F, *et al.* Artificial intelligence in cardiovascular medicine: clinical applications. *Eur Heart J* 2024;45:4291–304.
- [121] Dsw H, Schierding W, Wake M, *et al.* Machine learning SNP based prediction for precision medicine. *Front Genet* 2019;10:267.
- [122] Vilne B, Kibilds J, Siksa I, *et al.* Could artificial intelligence/machine learning and inclusion of diet-gut microbiome interactions improve disease risk prediction? Case study: coronary artery disease. *Front Microbiol* 2022;13:627892.
- [123] Oikonomou EK, Khera R. Machine learning in precision diabetes care and cardiovascular risk prediction. *Cardiovasc Diabetol* 2023;22:259.
- [124] Cheng X, Manandhar I, Aryal S, *et al.* Application of Artificial Intelligence in cardiovascular medicine. *Compr Physiol* 2021;11:2455–66.
- [125] Lip GYH, Genaidy A, Estes C. Cardiovascular disease (CVD) outcomes and associated risk factors in a medicare population without prior CVD history: an analysis using statistical and machine learning algorithms. *Intern Emerg Med* 2023;18:1373–83.
- [126] Yasmin F, Shah SMI, Naem A, *et al.* Artificial intelligence in the diagnosis and detection of heart failure: the past, present, and future. *Rev Cardiovasc Med* 2021;22:1095–113.
- [127] Alizadehsani R, Khosravi A, Roshanzamir M, *et al.* Coronary artery disease detection using artificial intelligence techniques: a survey of trends, geographical differences and diagnostic features 1991–2020. *Comput Biol Med* 2021;128:104095.
- [128] Velandia H, Pardo A, Vera MI, *et al.* Systematic review of Artificial Intelligence and electrocardiography for cardiovascular disease diagnosis. *Bioengineering* 2025;12:1248.
- [129] Siciliano GG, Onnis C, Barr J, *et al.* Artificial Intelligence applications in cardiac CT imaging for ischemic disease assessment. *Echocardiography* 2025;42:e70098.
- [130] Lin A, Kolossváry M, Motwani M, *et al.* Artificial intelligence in cardiovascular CT: current status and future implications. *J Cardiovasc Comput Tomogr* 2021;15:462–69.
- [131] Xu -P-P, Liu T-Y, Zhou F, *et al.* Artificial intelligence in coronary computed tomography angiography. *Med Plus* 2024;1:100001.
- [132] Matta S, Lamard M, Zhang P, *et al.* A systematic review of generalization research in medical image classification. *Comput Biol Med* 2024;183:109256.
- [133] Wessler BS, Nelson J, Park JG, *et al.* External validations of cardiovascular clinical prediction models: a large-scale review of the literature. *Circulation* 2021;144:e007858.
- [134] Edpuganti S, Shamim A, Gangolli VH, *et al.* Artificial Intelligence in cardiovascular imaging: current landscape, clinical impact, and future directions. *Discoveries (Craiova)* 2025;13:e211.
- [135] Biondi-Zoccai G, D'Ascenzo F, Giordano S, *et al.* Artificial Intelligence in cardiology: general perspectives and focus on interventional cardiology. *Anatolian J Cardiol* 2025;29:152–63.
- [136] Mastrodicasa D, van Assen M. Artificial intelligence for cardiac imaging is ready for widespread clinical use: pro Con debate AI for cardiac imaging. *BJR Open* 2025;7.
- [137] Tomala M. How algorithms are transforming the diagnosis of ischemic heart disease – state of the art. *Explor Cardiol* 2025;3:101275.
- [138] Lopez-Jimenez F, Attia Z, Arruda-Olson AM, *et al.* editors. Artificial intelligence in cardiology: present and future. *Mayo Clinic Proceedings*; 2020: Elsevier.
- [139] Karatzis L, Aung N, Aksentijevic D. Artificial intelligence in cardiology: hope for the future and power for the present. *Front Cardiovasc Med* 2022;9:945726.